2025/09 PHY301 Thermodynamic and Statistical Physics

Problems set 2

Lecturer: Song Kok Wee, kokwee.song@xmu.edu.my

Problem 1: Classical harmonic oscillator in Canonical ensemble [25 points]

Consider N harmonic oscillators with coordinates and momenta $\{q_i, p_i\}$ and subject to a Hamiltonian

$$\mathcal{H}(\{q_i, p_i\}) = \sum_{i=1}^{N} \left[\frac{p_i^2}{2m} + \frac{m\omega^2 q_i^2}{2} \right]$$

(a) In canonical ensemble, calculate the partition function $\mathcal{Z}$ for the classical osicillator using

$$\mathcal{Z} = \int \frac{d^{3N}q d^{3N}p}{h^3} \mathrm{e}^{-\beta\mathcal{H}}$$

Similar to the previous assignment, the integration is easier to carry out after and appropiate transformation.

(b) Using (a), calculate the Helmholtz free energy F, as a function of total energy E.

(c) Calculate the entropy S, and heat capacity C.

(d) Find the joint probability density $P(p, q)$ for a single oscillator. Hence calculate the mean kinetic energy, and mean potential energy for each oscillator.

Problem 2: Quantum harmonic oscillators in Canonical ensemble [25 points]

Consider N independent quantum oscillators subjects to a Hamiltonian

$$\mathcal{H}(\{n_i\}) = \sum_{i=1}^{N} \hbar\omega \left(n_i + \frac{1}{2} \right),$$

where $n_i = 0, 1, 2, \ldots$ is quantum occupation number for the ith oscillator.

(a) Using canonical ensemble, calculate the partition function $\mathcal{Z}$, as a function of temperature T.

(b) Calculate the energy E, and the heat capacity C, as functions of temperature T, and N.

(c) Find the probability $p(n)$ that a particular oscillator is in its nth quantum level.

Problem 3: Grand Canonical ensemble for relativistic gas [25 points]

Gas particles can adsorb on the surface of a solid at N possible adsorption sites. Each site has binding energy ϵ, and can accommodate at most one particle. The adsorbed particles are

in equilibrium with gas surrounding the solid. We can treat the system of adsorbed particles in a grand canonical ensemble with temperature T and chemical potential μ.

(a) If there are M adsorbed particles, what is the energy $E(M)$ of the system? What is the degeneracy $\Gamma(M)$ of the energy?

(b) Write down the grand partition function of the system as a sum over M. Determine the thermal average $\bar{M}$ as the value that maximized the summand in the grand partition function.

(c) Suppose the massless relativistic gas with Hamiltonian $\mathcal{H} = \sum_{i=0}^{N} p_i c$ surrounding the solid has pressure P. Calculate $\bar{M}$ using the ideal-gas expression for μ.

(d) Find $\overline{M^2} - \overline{M}^2$.

Problem 4: Lattice gas model as an Ising model with nonzero H [25 points]

Considering a 2D square lattice with N sites, each site can only occupy a particle and the occupation energy is ϵ. This gives the non-interacting Hamiltonian

$$\mathcal{H}_0 = \sum_{i=1}^{N} \epsilon n_i$$

where $n_i = 0, 1$ is the occupation number for i-th site. In the lattice gas model, the particle in the lattice has attractive interaction with it adjacent particles. This yields the interacting Hamiltonian as

$$\mathcal{H}_{\text{int}} = -\frac{1}{2} \sum_{\langle ij \rangle} u n_i n_j$$

Therefore, the total Hamiltonian is

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{\text{int}} = -\frac{1}{2} \sum_{\langle ij \rangle} u n_i n_j + \sum_{i=1}^{N} \epsilon n_i$$

(a) Express the occupation number n_i in term of spin $\sigma_i = \pm 1$. Rewrite the Hamiltonian $\mathcal{H}$ in term of spin σ_i as

$$\mathcal{H} = -\frac{1}{2} \sum_{\langle ij \rangle} J \sigma_i \sigma_j - H \sum_i \sigma_i$$

what is the relation between J, H and u, ϵ?

(b) In the mean-field method, we assume the fluctuation of the spin $\delta\sigma_i = \sigma_i - \langle \sigma_i \rangle$ is small where $\langle \sigma_i \rangle = m$ is a average spin on site i. Neglecting $\delta\sigma_i \delta\sigma_j$, we approximate $\mathcal{H} \approx \mathcal{H}_{MF}$ to a non-interacting spin system. Using $\mathcal{H}_{MF}$, calculate the partition function of a canonical ensemble.

(c) Show that the self-consistent for $\langle \sigma_i \rangle = m$ is

$$m = \tanh\left[4\beta J\left(m + \frac{H}{4J}\right)\right]$$

(d) Find the transition temperature T_C and m near T_C. What is the average occupation number $\langle n_i \rangle$ near T_C?

(e) At $H = 0$, considering m is small near T_C, expand the Helmholtz free energy F up m^4. Express F in term of the mean occupation $n = \langle n_i \rangle$.